

PRIYANSHU

priyanshu4096@gmail.com | 7986534542 | Nangal Township, Ropar, Punjab

[LinkedIn](#) | [GitHub](#) | [X \(Twitter\)](#) | [LeetCode](#)

SUMMARY

Software Engineering student with experience building AI-powered full-stack applications using MERN, LangChain, LangGraph, Pinecone, and Google Gemini. Interested in backend engineering, distributed systems, and applied AI with a strong foundation in Data Structures & Algorithms.

EDUCATION

B.Tech Computer Science & Engineering <i>Guru Nanak Dev University, Amritsar</i>	07/2023 – 2027 CGPA: 7.80
Class 12th CBSE <i>BBMB DAV Public School, Nangal</i>	04/2021 – 03/2022 83.6%
Class 10th CBSE <i>BBMB DAV Public School, Nangal</i>	04/2019 – 03/2020 85.33%

KEY SKILLS

Programming C++, JavaScript

Languages

Frameworks React.js, Node.js, Express.js

GenAI LangChain, LangGraph, Gemini, Pinecone, RAG

Database MongoDB, Redis, Pinecone

Tools Google Gemini AI (Embeddings & LLM), RAG Pipelines

Core Git, GitHub, Postman, VS Code, Cloudinary

PROJECTS

NovaCode – AI-Powered Coding Platform [Github](#) **MERN | Redis | Judge0 | Gemini AI**

- Architected a scalable full-stack online coding platform with **AI-assisted learning** and **real-time code evaluation**.
- Engineered secure authentication using **JWT**, **Google OAuth**, **bcrypt**, and **Redis-backed session management** for secure logout.
- Built a **RESTful backend** using **Node.js**, **Express.js**, and **MongoDB**, following modular architecture and scalable API design.
- Integrated the **Judge0 API** to support multi-language code execution (**C++**, **Java**, **JavaScript**) with real-time compilation and test case validation.
- Implemented **Redis-based rate limiting** to prevent API abuse and improve platform security and reliability.
- Built an **admin portal** for end-to-end problem lifecycle management, including CRUD operations, test cases, and uploading editorial/video solutions.
- Designed a dynamic problem management system featuring filtering by **difficulty**, **topic/tags**, and **submission status** for improved problem discovery.
- Leveraged **Google Gemini AI** to deliver intelligent coding hints and personalized **DSA guidance**.
- Integrated **Cloudinary** for secure storage and delivery of user-uploaded assets, including profile resumes and editorial videos.

DocuMind AI – Agentic Document Platform [Github](#) **LangChain | GeminiAI | Clerk | Pinecone | Redis**

- Implemented an asynchronous **PDF ingestion pipeline** that extracts text, generates **embeddings**, and indexes document chunks of upto 10mb into **Pinecone** in real time.
- Engineered an **agentic RAG pipeline** using **LangGraph** that dynamically validates retrieved context before generating responses, enabling multi-turn conversational reasoning.
- Built a semantic retrieval system using **768-dimensional embeddings** in **Pinecone**, supporting fast similarity search across large document collections.
- Developed a **multi-user SaaS architecture** with **Clerk authentication**, isolated document storage, and persistent conversation history.
- Implemented **Redis-backed rate limiting** to restrict users to **5 messages per 6 hours**, preventing API abuse and controlling AI inference usage.

ACHIEVEMENTS

- Solved **300+ coding problems** and demonstrated consistent problem-solving performance across Data Structures and Algorithms.